UNIT - IV

Infrared & Thermal Testing

1. Introduction and Fundamentals

* Infrared & Thermal Testing (IRT) is an NDT method that uses temperature differences and thermal radiation of a surface to find flaws, hidden defects and changes in the condition of a component.
* Principle: a defect changes the flow of heat through a part; this makes a local hot or cold spot on the surface, which is detected by contact thermometers, heat sensitive coatings or an infrared camera .
* Everything above absolute zero (0 K = -273 deg C) emits infrared radiation ; hotter objects emit more.
* Used for: delamination, disbonds, cracks, voids, corrosion, moisture, blocked flow, loose electrical joints, insulation faults, thickness loss.

Basic Terms

* Temperature (T): measure of thermal state - K or deg C.
* Heat (Q): energy transfer due to temperature difference - Joule.
* Thermal conductivity (k): W / m K - ability to conduct heat.
* Specific heat (c): energy to raise 1 kg by 1 K (J / kg K).
* Thermal diffusivity (alpha) = k / (rho c) - m^2 / s - how fast heat spreads. A low-diffusivity region (a void) holds heat.
* Thermal effusivity (e) = sqrt(k rho c) - governs contact temperature of two bodies.
* Thermal resistance R = L / (k A) .

2. Infrared Radiation - Electromagnetic Spectrum

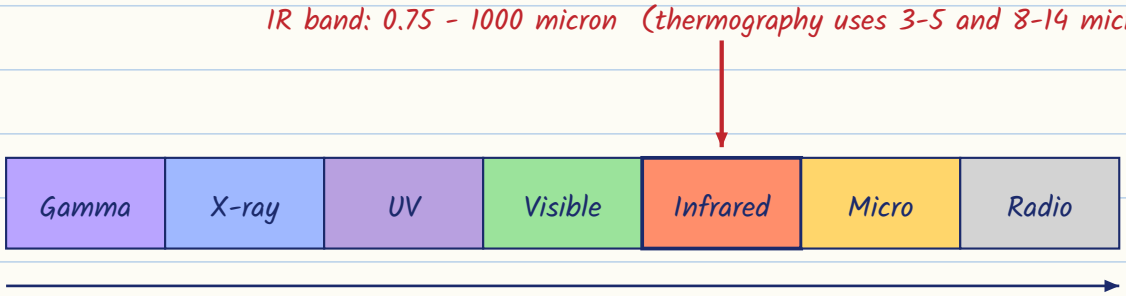


Fig: Infrared band in the electromagnetic spectrum

* Infrared region is between **visible red light (0.75 micron) and microwaves (1000 micron)** .
* Sub-bands: **Near IR (0.75 - 1.4 micron), Short wave IR (1.4 - 3), Mid wave IR (3 - 5 micron), Long wave IR (8 - 14 micron), Far IR (above 15 micron)** .
* Atmosphere is transparent only in windows **3 - 5 micron and 8 - 14 micron** , hence thermal cameras work in these bands.
* Infrared travels at light speed, in straight lines, can be reflected, refracted and absorbed.

Laws of Thermal Radiation

* **Black body:** ideal body that absorbs all radiation and emits the maximum possible at its temperature (emissivity = 1).
* **Stefan-Boltzmann law:** total energy radiated **W = epsilon sigma T^4** (sigma = 5.67 x 10^-8 W / m^2 K^4; epsilon = emissivity). Radiation increases as the **fourth power** of temperature.
* **Wien's displacement law: lambda max x T = 2898 micron K** - the peak wavelength shifts to shorter wavelength as T rises. For a body at 300 K, lambda max is about 9.7 micron (in LWIR).
* **Planck's law:** distribution of radiation over wavelength.
* **Kirchhoff's law:** good absorbers are good emitters, **absorptivity = emissivity** .

* **Emissivity (epsilon):** ratio of radiation of a real surface to that of a black body at the same temperature (0 to 1). Polished metal = 0.05 – 0.1; oxidised steel = 0.8; paint = 0.9; human skin = 0.98.
* Radiation seen by camera = **emitted + reflected + transmitted** part: **epsilon + rho + tau = 1**.

Remember:

Low emissivity shiny surface -> false low temperature reading. Cover with black paint or tape (known emissivity) or correct the emissivity setting.

3. Heat Transfer

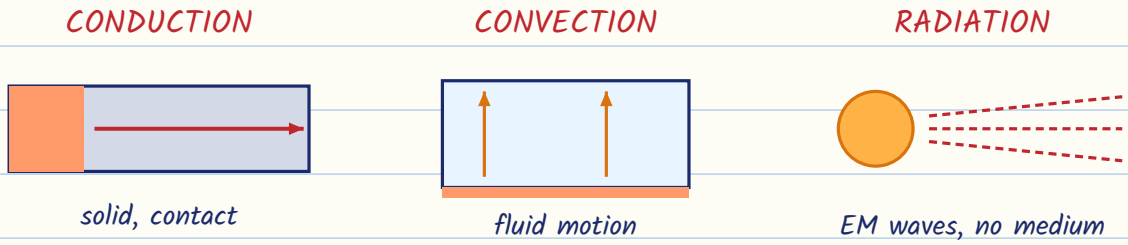


Fig: Modes of heat transfer

* **Conduction:** heat flow through a solid or between touching bodies due to temperature gradient. **Fourier's law: q = - k A dT / dx**. A flaw (air gap) has lower k and **obstructs** heat flow.
* **Convection:** heat transfer by motion of a fluid (liquid / gas). Newton's law of cooling: **q = h A (Ts - Tf)** ; natural or forced.
* **Radiation:** heat transfer by electromagnetic waves, needs no medium. **Net exchange q = epsilon sigma A (T1^4 - T2^4)**.
* In thermal NDT, **conduction inside** the part is related to flaw location and **convection and radiation at the surface** are the loss paths and the source of the signal.
* Defects:

- **Insulating** flaw (void, delamination) -> heat blocked -> **hot spot on heated side**, cooler surface behind it.
- **Conductive** flaw (metal inclusion in plastic) -> **cold spot** on heated side.
- Moisture and corrosion change heat capacity and conductivity.

4. Active and Passive Techniques

4.1 Passive Thermography

* The component is **naturally at a different temperature from its surroundings** due to operation or process. No external heating needed.
* **Examples:** hot bearings, overheated electrical joints and fuse, steam line insulation, furnace refractory, building heat loss, hot spots in PCB, moisture in roofs, human fever screening.
* Fast, simple, done on a **live equipment**, qualitative.

4.2 Active Thermography

* An **external stimulus (heating or cooling) is applied** to create a thermal contrast over defects.
* **Heat sources:** flash lamp, halogen lamp, laser, hot air, hot water, induction heating, ultrasound (vibro-thermography), microwave, Joule heating by current.
* **Modes of heating:** reflection mode (camera and source on same side) for sub-surface flaws; transmission mode (camera on opposite side) for through-thickness flaws.
* Gives **quantitative** depth and size of defects.

Point	Passive	Active
External energy	Not needed	Required (heat source)
Target	Operating equipment, in-service	Material defects, composites
Result	Qualitative, temperature anomaly	Quantitative defect depth / size
Example	Electrical panel check	Delamination of aircraft panel

5. Lock-in and Pulse Thermography

5.1 Pulse Thermography (PT)

* A short, high-energy heat pulse (flash 1 ms - few seconds) heats the surface; the IR camera records surface temperature decay with time.
* The heat diffuses inward; an insulating flaw at depth z reduces cooling rate -> appears as a warmer area after time t = z^2 / alpha (alpha = diffusivity).
* Deeper flaws appear later and with lower contrast. Gives depth from time of appearance.
* Advantages: fast (seconds), wide area, simple. Limitations: non-uniform heating, surface emissivity variation, and low depth resolution; needs high power flash.
* Processing: thermal signal reconstruction (TSR) , pulsed phase thermography (PPT) by Fourier transform.

5.2 Lock-in (Modulated) Thermography

* The surface is heated by a sinusoidally modulated source at frequency f_lock; the camera records the response at the same frequency ("locked-in").
* The amplitude and phase images are calculated; phase is independent of surface emissivity and non-uniform heating -> better detection of deep defects.
* Thermal diffusion length: mu = sqrt(alpha / (pi f)) - depth probed; lower frequency gives deeper penetration , up to about 1.8 mu depth in phase image.
* Advantages: high sensitivity, low power source, quantitative depth, deals with emissivity problem. Limitations: slower (needs several cycles), one frequency at a time.
* Variants: vibro-thermography / lock-in ultrasound.

Point	Pulse	Lock-in
Heating	Single short pulse	Periodic sine wave
Time	Seconds	Minutes
Image	Temperature vs time	Amplitude and phase

Depth range	Medium	Deeper, better phase
Equipment	Flash lamp, camera	Lamp / laser with modulator, camera

6. Contact and Non-contact Thermal Inspection Methods

6.1 Contact Methods (touch the object)

* **Thermometers:** mercury / alcohol type, bimetallic thermometers.
* **Thermocouples:** two dissimilar metals (K type chromel-alumel, J, T) produce emf (Seebeck effect); range -200 to 1300 deg C.
* **Resistance temperature detectors (RTD, platinum PT100)** - accurate, linear.
* **Thermistors** - semiconductor, sensitive but narrow range.
* **Heat sensitive paints, crayons, pellets, labels, papers, and liquid crystals** (below).
* **Limitations:** point measurement only, thermal contact error, needs access to the surface, slow response, disturbs the temperature of small objects.

6.2 Non-contact Methods

* **Radiation pyrometers / spot IR thermometers** - single point, quick, safe on moving, hot or live objects.
* **Optical pyrometer (disappearing filament)** for very hot bodies.
* **Thermal imaging (IR cameras)** - full-field 2-D temperature map.
* **Advantages:** no contact, fast, safe distance, area view. **Limitations:** depends on emissivity, reflections, atmosphere, distance.

7. Heat Sensitive Coatings and Materials

7.1 Heat Sensitive Paints

* Special paint that **changes colour at a fixed temperature** (irreversible, indicates maximum temperature reached). Each paint has a transition temperature range (about 40 to 1300 deg C). Applied by brush or spray.
* Used on **turbine blades, engine parts, electrical bus bars, furnace** , boilers.
* Types: single change and **multi-change (several colours)** paints; **thermo-chromic** reversible paints.

7.2 Heat Sensitive Papers

* Papers impregnated with chemicals (such as ones that change from white to black) at a certain temperature; used as ***temperature labels / strips*** and in thermal printers. Quick checks on parts in ***inaccessible*** locations.

7.3 Thermally Quenched Phosphors

* ***Phosphors*** glow under UV light; as the ***temperature rises, the brightness of the luminescence goes down*** ("thermal quenching") and the decay time shortens.
* A phosphor coating (such as magnesium fluorogermanate) is applied on the surface, excited with UV; the ***brightness map*** shows the temperature distribution. Used for ***non-contact surface temperature mapping*** of electronic circuit boards, turbine blades.

7.4 Liquid Crystals

* ***Organic compounds*** that exist between solid and liquid phase and ***selectively reflect light of a colour depending on temperature*** - they change colour from ***red -> yellow -> green -> blue -> violet*** with rise in temperature.
* ***Cholesteric (chiral nematic) liquid crystals*** are used; range of colour play can be fixed from -30 to 250 deg C with very narrow band 1 - 2 deg C accuracy.
* They are ***reversible*** , give full surface display and need no electronic equipment.
* ***Applications:*** detecting hot spots on printed circuit boards, bonded (adhesive) joints, voids in honeycomb, cracks in welds, delamination, medical thermography (skin temperature).

7.5 Techniques for Applying Liquid Crystals

* Surface must be cleaned and coated with ***black paint (background)*** for good colour contrast.
* ***Spraying:*** liquid crystal mixture in solvent sprayed with an airbrush on the black surface.
* ***Brush / painting*** on a flat surface.
* ***Encapsulated (micro-encapsulated) slurry or sheet:*** the crystals are in tiny capsules in a binder and applied by spraying or as ***self-adhesive sheets*** - easy to use, durable and protected from contamination.

* **Dip coating, printing** of thin films for tests of small parts.
* Viewing in white light; record with a colour camera.

7.6 Other Temperature Sensitive Coatings

* **Thermographic phosphors, temperature indicating crayons (Tempilstik),** thermochromic inks, **pyrometric cones, fusible pellets and melting point paints** (melt at exact temperature – 40 to 1400 deg C).
* **Thermal barrier coatings** check; **pressure and temperature sensitive paints** in wind tunnel tests.

8. Inspection Methods Using Infrared

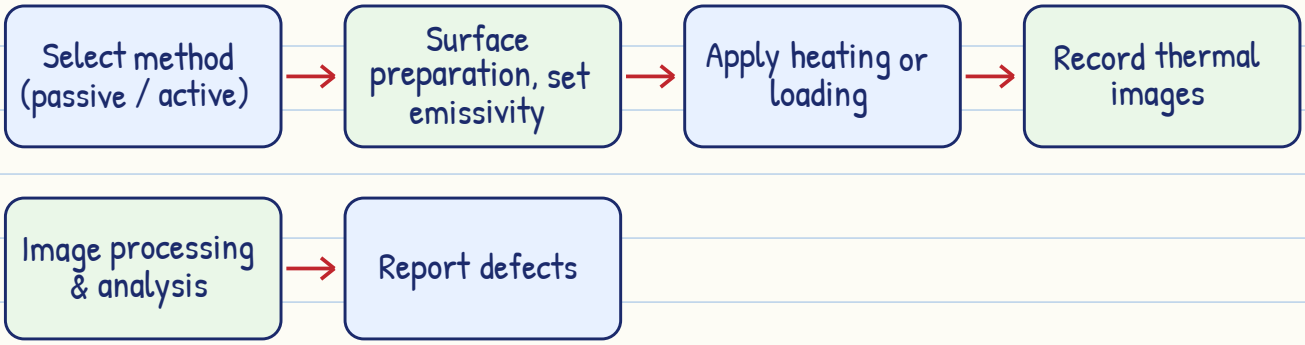


Fig: Procedure for IR thermographic inspection

* **Surface preparation:** clean surface, apply thin black matte paint if surface is shiny (to increase emissivity and avoid reflections).
* **Calibration:** camera with a **black body reference source** ; set emissivity, reflected apparent temperature, distance, humidity.
* **Select view angle and camera distance** (spatial resolution IFOV).
* **Stimulation:** thermal (flash, lamp, hot air), mechanical (ultrasound / vibration), electrical (current), or microwave / eddy current.
* **Record sequence** at suitable frame rate and duration (some seconds to minutes).
* **Image processing:** background subtraction, contrast enhancement, FFT, principal component analysis, TSR.
* **Interpretation:** compare with reference area, find the defect size, depth; report with thermogram, visual image, settings, date, ambient conditions.

9. Infrared Radiation and Infrared Detectors

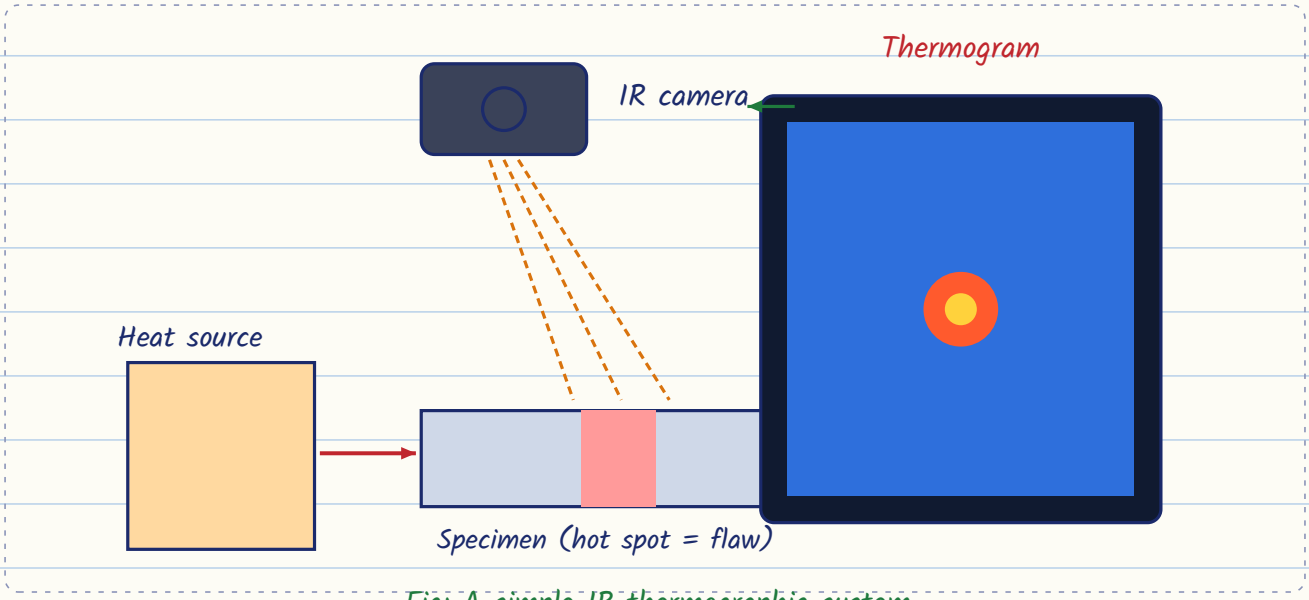


Fig: A simple IR thermographic system

9.1 Infrared Camera System

* **Optics (germanium lens)** - IR transparent materials: Ge, ZnSe, silicon, sapphire (glass does not pass LWIR).
* **Detector (focal plane array)** converts IR radiation into electrical signal.
* **Cooling system** (Stirling or liquid nitrogen for cooled detectors).
* **Signal processor, display, memory** and software.

9.2 Types of IR Detectors

* **Thermal detectors:** respond to **heating effect** of radiation; wavelength independent, uncooled, slower, cheaper.
  - **Bolometers** (microbolometer array of vanadium oxide / amorphous silicon) - resistance changes with temperature; most common.
  - **Thermocouples / thermopiles** , **pyroelectric detectors (TGS, lithium tantalate)** - charge changes with temperature.
* **Photon (quantum) detectors:** radiation frees electrons directly in semiconductor; **very sensitive and fast** , wavelength dependent; usually **cooled (77 K)** to reduce noise.
  - **InSb (indium antimonide, 3 - 5 micron), HgCdTe (MCT, 3 - 12 micron), QWIP (quantum well, 8 - 9 micron), PbS, PbSe, InGaAs** .

Point	Thermal detector	Photon detector
Response	Heat effect	Photon absorption
Speed	Slow (ms)	Fast (microseconds)
Sensitivity	Moderate (NETD 50 mK)	High (NETD below 20 mK)
Cooling	Not required	Required (cryogenic)
Cost	Low	High

* Performance terms: NETD (noise equivalent temperature difference), spatial resolution (IFOV), field of view, frame rate, spectral range, accuracy +/- 2 deg C.

10. Thermo-mechanical Behaviour of Materials

* Thermal expansion: a body expands when heated, change in length dL = alpha_L L dT (alpha_L = coefficient of linear expansion).
* When expansion is constrained, thermal stress = E alpha_L dT develops. Uneven heating -> thermal gradient -> stress, warping, cracks, fatigue (thermal cycling).
* Thermoelastic effect: when a body is elastically loaded (tension or compression), its temperature changes slightly - rise in compression and fall in tension. Thermoelastic stress analysis (TSA) uses IR camera to map the stress field: dT = - K T (sigma1 + sigma2), where K = thermoelastic constant.
* Heat generated in plastic deformation, friction and crack tip - the fatigue crack growth, bolt looseness and rubbing parts show heating detected with IR camera.
* Vibro-thermography (sonic IR): ultrasonic energy vibrates faces of a crack; friction at the crack faces generates heat so only closed cracks and delaminations appear hot.
* Creep, thermal fatigue and thermal shock are detected by monitoring temperature of boiler, turbine parts.
* Material property and heat treatment (conductivity, diffusivity) can be mapped by thermography.

11. IR Imaging Applications

11.1 Aerospace Applications

* **Composite structures (CFRP, GFRP):** detection of **delamination, disbonds, impact damage (BVID), porosity, water ingress** in wings, fuselage panels, rudders, radomes.
* **Aircraft skin:** corrosion and lap-joint disbonds, thinning, rivets; repair patch quality.
* **Engine and turbine blades:** cooling hole blockage, thermal barrier coating (TBC) delamination, cracks; space shuttle tiles / heat shield inspection.
* **Rocket and missile:** motor case bond line, propellant voids, insulation.
* Pulse thermography and lock-in used for fast checking of large panels.

11.2 Electronic Components

* **Hot spot detection** on PCB - short circuit, overloaded or failing component, poor solder joint, delamination of chips, heat sink contact.
* **Thermal design verification** of components and power devices (IGBT, LED); **failure analysis** of integrated circuits (lock-in thermography finds micro leakage).
* Switchgear, transformers, bus bars, connections and cables in **electrical substations** (predictive maintenance).

11.3 Honeycomb and Sandwich Structures

* Sandwich panels have thin **skins bonded to a honeycomb core** (Al, Nomex, foam).
* Defects: **disbond (skin-core), crushed core, water ingress, node failure, adhesive voids** .
* Pulse or lock-in thermography with transmission or reflection mode: **disbonded area retains heat and appears hot** , water-filled cells appear cooler on cooling due to high heat capacity.
* Liquid crystals also used for quick checks of honeycomb bond.

11.4 Other Applications

* Buildings (insulation and moisture), pipelines, refinery furnaces, boilers tubes, refractory lining, welding monitoring, medical thermography, forest fire, mines.

12. Case Studies

Case 1 - Delamination in a CFRP Aircraft Panel

* **Problem:** impact damage on carbon fibre wing skin, invisible on surface.
* **Method:** active pulse thermography using two flash lamps (2 ms, 6 kJ), IR camera at 100 Hz, reflection mode.
* **Result:** delamination region showed **slower cooling (warmer)** at t = 0.4 s; size and depth estimated by TSR. **Panel repaired** after the area was marked.

Case 2 - Hot Joint in an Electrical Panel

* **Problem:** periodic trips in a 415 V distribution panel.
* **Method:** passive IR survey under load.
* **Result:** a loose lug on one phase was **35 deg C above** the other phases -> torque corrected and temperature normal. Avoided fire and downtime.

Case 3 - Honeycomb Disbond in a Helicopter Blade

* **Method:** lock-in thermography at 0.1 Hz with halogen lamps.
* **Result:** phase image displayed the **skin-core disbond** as clear patches; confirmed by tap test and ultrasonic. Early detection avoided blade failure.

Case 4 - PCB Failure Analysis

* **Method:** IR camera on powered board; one capacitor showed **60 deg C hot spot** ; found a short circuit due to solder bridging.

13. Advantages and Limitations of IR Thermography

* **Advantages: non-contact** , safe, fast, full-field image, works on live equipment, remote and hazardous places, sensitive, easy to record and report, no radiation.
* **Limitations: surface temperature only** (limited depth); depends on emissivity and reflections; shiny, low emissivity metals are difficult; thick or high conductivity materials; cost of high-quality cameras; need trained thermographer; effect of wind, sun, moisture, ambient temperature; the results are affected by thermal contrast and heat sources.

Summary:

IRT detects flaws from surface temperature patterns. Passive for operating equipment, active (pulse / lock-in) for defects. Needs correct emissivity, IR camera in 3 - 5 or 8 - 14 micron window. Used widely on composites, electronics and sandwich panels.

Important Questions

* Explain the principle of thermal NDT and the laws of thermal radiation.
* Differentiate active and passive thermography; explain pulse and lock-in techniques.
* Describe heat sensitive paints, liquid crystals and thermally quenched phosphors.
* Explain IR detectors and the working of an IR camera.
* Write notes on IR imaging in aerospace and electronic components with a case study.